Q6: Text Generation Using OpenAI API

Aim: To develop a Python program that connects to an OpenAI Large Language Model (LLM) using the OpenAI API and generates text based on a user-provided prompt.

Algorithm

1. Start the program.
2. Install and import the OpenAI library.
3. Enter the OpenAI API key securely.
4. Create an OpenAI client.
5. Get a text prompt from the user.
6. Send the prompt to the OpenAI model.
7. Display the generated text.
8. Stop the program.

```python
from transformers import pipeline

# Load a free, pre-trained text-generation model
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

# Get input from the user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=80,
    do_sample=True,
    temperature=0.7,
    pad_token_id=generator.tokenizer.eos_token_id
)

# Display generated text
print("\nGenerated Text:")
print(result[0]["generated_text"])
```

```
Loading weights: 100% 76/76 [00:00<00:00, 2904.96it/s]
Enter your prompt: Explain Artificial Intelligence in simple words

Generated Text:
Explain Artificial Intelligence in simple words, and I think you'll love it.


This post is about the artificial intelligence and the artificial intelligence.
It's about the potential for artificial intelligence to be made better and more effective by the use of artificial intelligence.
The AI could be a good thing for us to understand more about human behavior. (That's why I'm writing this post about artificial intelligence.)
In fact
```

Result

Thus, a Python program was developed to connect to the OpenAI API and generate text based on a user-provided prompt.

Q7: Text Generation Using Gemini API

Aim: To develop a Python program that connects to the Google Gemini API and generates text based on a user-provided prompt.

Algorithm

1. Start the program.
2. Install and import the Google GenAI library.
3. Enter the Gemini API key.
4. Create the Gemini client.
5. Get a prompt from the user.
6. Send the prompt to the Gemini model.
7. Display the generated text.
8. Stop the program.

code:

```
from transformers import pipeline

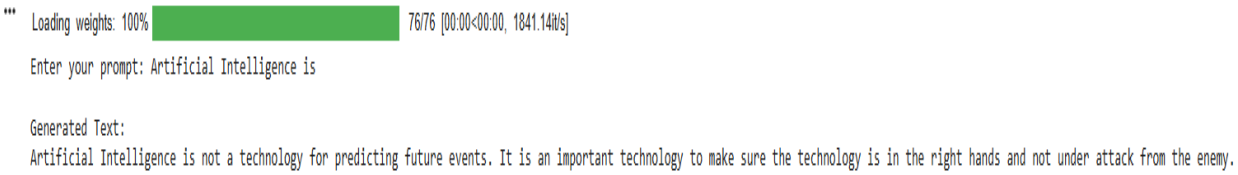
# Load a pre-trained language model
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

# Get prompt from user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=60,
    do_sample=True,
    temperature=0.7,
    pad_token_id=generator.tokenizer.eos_token_id
)

# Display output
print("\nGenerated Text:")
print(result[0]["generated_text"])
```

OUTPUT:



Result: Text generation was performed using a pre-trained language model in Google Colab without requiring an API key.